Experiment 20

Pandas Library Use Cases in Data Science.

Program:

```
import numpy as np
import pandas as pd

list=[[1,'Smith',50000],[2,'Jones',60000]]
df=pd.DataFrame(list)
df.columns=['Empd','Name','Salary']
df
df.info()

df=pd.read_csv("/content/50_Startups.csv")
df.info()
df.head()
df.tail()

df=pd.read_csv("/content/employee.csv")
df.head()
df.tail()
df.info()

df.salary
type(df.salary)
df.salary.mean()
df.salary.median()
df.salary.mode()
df.salary.var()
df.salary.std()
df.describe()
df.describe(include='all')

empCol=df.columns
empCol
emparray=df.values
employee_DF=pd.DataFrame(emparray,columns=empCol)
employee_DF
```

Output

```
Employee DataFrame output includes:
   Empd / Name / Salary
   1 Smith 50000
   2 Jones 60000

Employee salary statistics:
   mean     5714.285714285715
   median   6000.0
   mode     5000, 6000
   var      2571428.5714285714
   std      1603.5674514745463

Final employee_DF:
   1 SREE VARSSINI K S       5000
   2 SREEMATHI B             6000
   3 SREYA G                 7000
   4 SREYASKARI MULLAPUDI    5000
   5 SRI AKASH U G           8000
   6 SRI HARSHAVARDHANAN R   3000
   7 SRI HARSHAVARDHANAN R   6000
```